Physical momentum, energy, and the remaining quantum frame obstruction in the Toda–Boussinesq corner

Quantum Boussinesq solitons project

4 November 2026

Abstract

(c) We derive an energy counterpart of the endpoint momentum identity, relate the endpoint responses to derivatives of dressed momentum and energy, and give an independently implemented finite-sea dictionary and numerical energy dispersion at fixed dressed momentum. A mixed rectangle of the classical-frame symmetric remainder stays nonzero under the tested edge extrapolations. These are consequences of the existing asymptotic Bethe integral equations, not a new proof of their microscopic applicability. **(d)** The requested physical scalar one-loop scattering prediction is not determined: the endpoint identities leave an antisymmetric two-body frame function free. We exhibit an explicit freedom that survives the nested limit while leaving every one-body observable and reciprocity test intact. This is a checked partial result, not a completed common quantum dictionary. No Wiener–Hopf dispersion or phase formula from Phase 2i is assumed.

1 Scope, conventions, and input status

Labels (a), (b), (c), (d) have the repository meanings: literature proof, literature heuristic, project derivation/check, and guess or unresolved identification. Numerical extrapolations labelled (c) below are explicitly numerical evidence; they are not proofs of existence of the limiting functions. All new formulas are checked by `frame_identities.py`, `frame_edge.py`, or `frame_analysis.py`, as indicated by section. No external source is cited or claimed to have been read. The full texts in `literature/` were unavailable. The inputs read were the master report (Route D and ranked priorities), `toda_route.tex`, `toda_edge.tex`, `matching.tex`, the Toda sections of `charge_frame_master.tex`, `independent_phase2h_audit.tex`, `audit_toda_dictionary.tex`, Section 4 of `audit_frames_defects.tex`, their specified implementations and saved JSON data, and the relevant continuum formulas in `oneloop_phase.tex`. The overstatements in earlier notes are not adopted here.

Inherited conventions. Use $h = \hbar_T$, $t = 2\pi h$, mean lattice spacing a, and $b = e^{-a/2}$. The physical coordinate of the unperturbed chain is $x_T = an$. The inherited leading classical dictionary is

$$\kappa = \sqrt{3}\beta, \quad h = 3b\hbar_B, \quad x_B = 2\sqrt{3}n, \quad t_B = 2\sqrt{3}bt_T, \qquad P_B = \frac{a\overline{P}}{6\sqrt{3}b}, \quad E_B = \frac{\overline{E}}{6\sqrt{3}b^2}. \tag{1}$$

(c) The charge scalings agree with the small-amplitude soliton expansions (`frame_identities.py`); they are leading continuum scalings, not exact finite-amplitude identities or a derivation of quantum collective charges. The bad-equation input is

$$u_{tt} = u_{xx} + u_{xxxx} + 3(u^2)_{xx}, \quad k = \sinh\beta, \quad v = \epsilon\cosh\beta,$$
$$P_{B,0} = \tfrac{2}{3}\epsilon\sinh^3\beta\cosh\beta, \qquad E_{B,0} = \tfrac{2}{15}\sinh^3\beta(5 + 4\sinh^2\beta).$$

The good convention instead has $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$, $k = \sin\alpha$, $v = \epsilon\cos\alpha$, $P_{g,0} = \frac{2}{3}\epsilon\sin^3\alpha\cos\alpha$. Both use $u = 2(\log\tau)_{xx}$ and the stipulated symplectic normalization. A larger right-moving bad/Toda soliton is faster; a smaller good soliton is faster. **(c)** This follows by differentiating the displayed velocities. **(d)** No real quantum equivalence between good and bad models, or analytic continuation of their quantum states, follows from this classical dictionary. We keep the inherited classical coordinate scale; we do not rescale time to set the quantum sound velocity to one.

2 An exact finite-sea one-body dictionary

Definitions and derivation (c) (`frame_identities.py`, `frame_edge.py`). For an invertible finite-sea operator on $[-q, q]$, write

$$\Theta(d) = 2h\,\mathrm{Im}\log\Gamma(1 + id/h) + 2d\log h, \qquad K(d) = \Theta'(d), \tag{2}$$

$$(\mathcal{A}f)(x) = tf(x) - \int_{-q}^{q} K(x-z)f(z)\,\mathrm{d}z, \qquad \mathcal{A}\rho = 1, \quad \mathcal{A}\eta = x, \quad \int\rho = 1/a, \tag{3}$$

$$\mathcal{A}F_p(x) = \Theta(p-x), \qquad P(p) = p + \int\Theta(p-x)\rho(x)\,\mathrm{d}x, \tag{4}$$

$$E(p) = \frac{p^2}{2} + \int xF_p(x)\,\mathrm{d}x = \frac{p^2}{2} + \int\Theta(p-x)\eta(x)\,\mathrm{d}x. \tag{5}$$

The last equality uses self-adjointness of $\mathcal{A}$. Thus only two sea solutions, ρ and η, are needed for all one-body observables. For the right edge define

$$\begin{aligned} \overline{P}(p) &= P(p) - P(q), & \overline{E}(p) &= E(p) - E(q), \\ D(p) &= 1 + \int K(p-x)\rho(x)\,\mathrm{d}x, & V(p) &= p + \int K(p-x)\eta(x)\,\mathrm{d}x. \end{aligned} \tag{6}$$

(c) Then $P' = D$, $E' = V$, $P(q) = \pi h/a$, $D(q) = t\rho(q)$ and $V(q) = t\eta(q)$. The edge-subtracted energy uses the existing zero-energy edge reference; no chemical-potential constant is fitted. Parity gives $P(-p) = -P(p)$ and $E(-p) = E(p)$, so the left-edge version follows with signed momentum. The excitation group velocity and right sound speed are

$$v_{\mathrm{exc}}(p) = \frac{V(p)}{D(p)}, \qquad v_s = \frac{\eta(q)}{\rho(q)}. \tag{7}$$

These are velocities conjugate to the dressed physical momentum, not the bare Bethe parameter p.

Integration by parts in the differentiated F_p equation gives

$$\boxed{t\rho(q)[F_p(q) - F_p(-q)] = 1 - D(p), \qquad t\eta(q)[F_p(q) + F_p(-q)] = P(p) - V(p).} \tag{8}$$

For example, multiply $\mathcal{A}\partial_x F_p$ by η and use $\int x\partial_x F_p = q[F_p(q) + F_p(-q)] - \int F_p$. The explicit endpoint terms cancel the bare q terms and leave $E'(p) = p + \int F_p - t\eta(q)[F_p(q) + F_p(-q)]$; self-adjointness gives $p + \int F_p = P(p)$. This proves the second identity without assuming a Wiener–Hopf factorization. **(c)**

Exact quantum Jacobian (c) (`frame_identities.py`, `frame_edge.py`). The old variable y is best understood as a monotone *physical-momentum label*, not a bare spectral coordinate:

$$\overline{P}(q + hu(y)) = P_0(\kappa) = \frac{4b}{a}(\kappa\cosh\kappa - \sinh\kappa), \qquad \kappa = \sqrt{yh/b}. \tag{9}$$

Consequently the exact finite-h dictionary is the parametric pair $(\overline{P}(q+hu), \overline{E}(q+hu))$, with

$$\boxed{\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{2\sinh\kappa}{aD(q+hu)}, \qquad \frac{\mathrm{d}\overline{E}}{\mathrm{d}y} = \frac{2h\sinh\kappa}{a}\frac{V(q+hu)}{D(q+hu)}.} \tag{10}$$

The classical comparator is $u_{0,h}(y) = 2b(\cosh\kappa - 1)/h$ and $u'_{0,h} = \sinh\kappa/\kappa$. Subtracting these known finite-amplitude expressions before extrapolating removes a large classical lattice artifact. All new calculations use the exact endpoint q, with no $q(1+10^{-10})$ offset. Fixed nonzero $\overline{P}$ as $h \to 0$ and fixed y are different limits: at fixed y, $a\sqrt{b}\,\overline{P}/h^{3/2} \to \frac{4}{3}y^{3/2}$. The term "fixed physical momenta in the edge limit" below means fixed *scaled* dressed momenta specified by (9).

3 Energy dispersion: what can be extracted without a phase frame

Classical comparison and loop order (c) (`frame_identities.py`). The classical energy is $E_0(\kappa) = 2b^2(\sinh\kappa\cosh\kappa - \kappa)$; $dE_0/dP_0 = ab\sinh\kappa/\kappa$. Moreover

$$E_0 - abP_0 = \frac{2b^2}{15}\kappa^5 + O(\kappa^7), \qquad E_{B,0} - P_{B,0} = \frac{1}{5}\beta^5 + O(\beta^7). \tag{11}$$

At fixed spectral coordinate, suppose locally away from the edge that $\overline{P} = P_0(p) + hP_1(p) + o(h)$ and $\overline{E} = E_0(p) + hE_1(p) + o(h)$. Inverting at fixed $P = P_0(p_0)$ gives

$$p = p_0 - h\frac{P_1(p_0)}{P_0'(p_0)} + o(h), \qquad E(P) = E_0(p_0) + h\left[E_1(p_0) - \frac{E_0'(p_0)}{P_0'(p_0)}P_1(p_0)\right] + o(h). \tag{12}$$

This is an exact formal chain-rule statement conditional on the expansions, not a uniform endpoint expansion. An $O(h)$ energy correction is the one-loop correction to the energy, whereas the one-loop *phase* is $O(h^0)$ after division of an action by h.

Independent numerical data (c) (`frame_edge.py`, `frame_analysis.py`). At $a = 1$ the calculation supplies a useful finite-y coefficient

$$c_{E,h}(y) = \frac{\overline{E}(P_0(\sqrt{yh/b})) - E_0(\sqrt{yh/b})}{hP_0(\sqrt{yh/b})}. \tag{13}$$

A finite extrapolated $c_E(y)$ would give $\overline{E} = E_0(P) + hPc_E(y) + o(hP)$ at fixed y. It contains the quantum crossover at that y; it is not itself a large-y loop expansion. Table 1 records the independent fits. In the inherited units (1) its leading quantum contribution to bad-Boussinesq energy is $3\hbar_B P_B c_E(y)$ for $a = 1$. **(d)** This is a Toda-induced small-amplitude dispersion, not a comparison with a fully renormalized continuum energy operator. Quantum corrections to the identification of collective charges remain a separate question.

(c) At fixed, unscaled momenta the independent finite-h extrapolation also reproduces the earlier dispersion estimates; Table 2 reports the coefficient $[\overline{E}(P) - E_0(P)]/h$. The sound-velocity estimate is tabulated separately and is not silently removed by a quantum change of coordinates.

(c) The extrapolated $(v_s - b)/h$ is 0.0795774724, with edge-fit spread $5.7\,10^{-9}$ and measured mesh difference $1.2\,10^{-11}$. This is a numerical estimate, not an independently proved closed form.

4 Endpoint responses: derivatives, not yet soliton charges

Response dictionary (c) (`frame_identities.py`, `frame_edge.py`). Put $f_p^{\pm} = F_p(\pm q) - F_q(\pm q)$ and use the earlier audit's responses

$$\mathfrak{M}_p = ta\rho(q)(f_p^+ - f_p^-), \qquad \mathfrak{Q}_p = \frac{f_p^+ + f_p^-}{2a\rho(q)}. \tag{14}$$

y	$u-y$	spread	$u\prime$	spread	$c_E(y)$	spread
0.1	-0.0763583	$6.9\,10^{-9}$	0.3524490	$9.6\,10^{-8}$	0.0725266	$5.9\,10^{-9}$
0.5	-0.2505729	$5.6\,10^{-8}$	0.7041319	$1.3\,10^{-7}$	0.0601062	$2.3\,10^{-8}$
1	-0.3574864	$1.3\,10^{-7}$	0.8429770	$1.8\,10^{-7}$	0.0559987	$7.3\,10^{-8}$
2	-0.4663924	$3.7\,10^{-7}$	0.9213452	$2.8\,10^{-7}$	0.0543111	$2.2\,10^{-7}$
4	-0.5757147	$1.1\,10^{-6}$	0.9604744	$4.4\,10^{-7}$	0.0536443	$7.2\,10^{-7}$
8	-0.6854957	$3.5\,10^{-6}$	0.9801739	$7.5\,10^{-7}$	0.0533400	$2.5\,10^{-6}$
16	-0.7955379	$1.2\,10^{-5}$	0.9900702	$1.3\,10^{-6}$	0.0531937	$8.8\,10^{-6}$
32	-0.9057144	$4.2\,10^{-5}$	0.9950310	$2.4\,10^{-6}$	0.0531202	$3.2\,10^{-5}$
64	-1.0159485	$1.5\,10^{-4}$	0.9975150	$4.2\,10^{-6}$	0.0530723	$1.1\,10^{-4}$

Table 1: Numerical edge extrapolations at fixed scaled dressed momentum (c). The spreads concern edge-fit models only. Maximum measured mesh differences across these rows are $5.1\,10^{-10}$ for $u-y$, $1.9\,10^{-11}$ for $u\prime$, and $4.7\,10^{-10}$ for c_E. Their propagated mesh diagnostics are at most fifteen times these values. No large-y asymptotic coefficient is fitted here.

P	$E_1(P)$	edge-fit spread	mesh difference
0.15	0.007775546	$7.5\,10^{-8}$	$8.9\,10^{-12}$
0.4	0.020319171	$3.7\,10^{-8}$	$3.9\,10^{-11}$
0.6	0.030098075	$2.2\,10^{-8}$	$7.1\,10^{-11}$

Table 2: One-loop energy at fixed, unscaled physical momentum (c, numerical). These are finite-amplitude Toda results, not finite-amplitude Boussinesq predictions.

The new energy identity turns these into directly measurable derivatives of the one-body Bethe dispersion:

$$\boxed{\mathfrak{M}_p = -a[D(p)-D(q)], \qquad \mathfrak{Q}_p = \frac{\overline{P}(p)-V(p)+V(q)}{2at\rho(q)\eta(q)}.} \tag{15}$$

There is also a thermodynamic interpretation of the *unsubtracted* sum. At fixed h vary q and let $n = 1/a = \int \rho$. Differentiating the sea problem and using self-adjointness yields

$$\partial_q n = 2t\rho(q)^2, \qquad \partial_q P(p) = t\rho(q)[F_p(q)+F_p(-q)], \qquad -a\partial_a P(p)\big|_{p,h} = \frac{F_p(q)+F_p(-q)}{2a\rho(q)}. \tag{16}$$

Thus $\mathfrak{Q}$ is a difference of these fixed-bare-parameter pressure responses, with the second derivative evaluated at $p=q$ *after* differentiation. It is not a derivative keeping $p-q$ fixed or following a fixed physical-momentum excitation. Those operations must not be conflated. The checks use independent finite differences in q as well as endpoint identities, at 60 decimal digits.

(d) Equations (15)–(16) do not identify $\mathfrak{M}$ with a quantum integrated compression or $\mathfrak{Q}$ with $P-\Pi$ in a chosen collective-coordinate Hilbert space. Their classical analogy alone is insufficient for that step.

5 Two-body references and the surviving mixed diagnostic

Exact decomposition (c) (`frame_identities.py`, `frame_edge.py`)**.** Keep the directional Bethe kernel

$$\Phi(p,z) = \Theta(p-z) - \int K(p-x)F_z(x)\,\mathrm{d}x. \tag{17}$$

Its four-term right-edge reference is $\widehat{\Phi}(p,z) = \Phi(p,z) - \Phi(p,q) - \Phi(q,z) + \Phi(q,q)$. The old two-term reference is $N(p,z) = \Phi(p,z) - \Phi(p,q) = \widehat{\Phi}(p,z) + U(z)$ with $U(z) = \Phi(q,z) - \Phi(q,q)$. Thus switching these references adds a one-body function. The exact endpoint reciprocity identity becomes

$$\widehat{\Phi}(p,z) + \widehat{\Phi}(z,p) = t(f_p^+ f_z^+ - f_p^- f_z^-) = \mathfrak{M}_p\mathfrak{Q}_z + \mathfrak{M}_z\mathfrak{Q}_p. \tag{18}$$

Our independent implementation checks it against the directly solved phases, not only against another expression in the same response functions. One algebraically antisymmetric representative is

$$B(p,z) = \widehat{\Phi}(p,z) - \mathfrak{M}_p \mathfrak{Q}_z. \tag{19}$$

(d) We do not select B/h as the physical scalar scattering phase. Antisymmetry is necessary for such a proposal, but is not its derivation.

A reference-free numerical test (c) (`frame_edge.py`, `frame_analysis.py`). At the fixed dressed momenta of (9) define

$$C_{\mathrm{cl}}(y,z) = -2\kappa_y[P_0(\kappa_z) - 2b\sinh\kappa_z], \tag{20}$$

$$S_h(y,z) = \frac{\mathfrak{M}_y\mathfrak{Q}_z + \mathfrak{M}_z\mathfrak{Q}_y - C_{\mathrm{cl}}(y,z) - C_{\mathrm{cl}}(z,y)}{h}, \tag{21}$$

$$\mathcal{D}_h(Y) = S_h(2Y,Y/2) - S_h(Y,Y/2) - S_h(2Y,Y/4) + S_h(Y,Y/4). \tag{22}$$

All calculations here use $a = 1$ and the exact quantum q. Every additive one-body reference disappears from $\mathcal{D}_h$. Its values therefore test genuine mixed dependence in the retained classical frame, and the same rectangle is obtained from the two-term normalization. Table 3 gives the new extrapolation.

Y	$\mathcal{D}_0(Y)$ estimate	edge-fit spread	mesh difference
4	-0.5043623	$3.7\,10^{-4}$	$6.0\,10^{-11}$
8	-0.4349827	$1.3\,10^{-3}$	$1.4\,10^{-10}$
16	-0.4087054	$5.2\,10^{-3}$	$2.4\,10^{-10}$
32	-0.3957520	$2.2\,10^{-2}$	$3.1\,10^{-10}$

Table 3: Mixed rectangles of the classical-frame symmetric discrepancy (c, numerical). The two largest momenta are in ratio $2^{3/2}$ at leading edge order; all four corners use separately inverted dressed momenta.

(c) Fitting these four edge estimates to $A + B/\sqrt{Y}$, $A + B/Y$, or $A + B/\sqrt{Y} + C/Y$ gives $A = -0.3276, -0.3774, -0.4034$, respectively. The fit-model range is about 0.0758, separately from the tabulated edge-fit errors. These unconstrained fits suggest a nonzero mixed asymptote but do not establish it; no fit is selected as the physical one-loop prediction.

(c) These data support survival of the nonseparable discrepancy at fixed scaled physical momenta as $h \to 0$, at the tested finite Y. This is stronger than the earlier single finite-h counterexample. **(d)** They do not prove a nonzero $Y \to \infty$ limit. Different large-Y fit models are reported in the JSON; extrapolation of four moderately large values is not an asymptotic theorem. Nor does a symmetric directional-kernel discrepancy disprove an ordered forward amplitude: it constrains a proposed common scalar frame.

Endpoint subtraction cannot be dropped. (c) In (15) the difference $D(p) - D(q)$ is essential. Replacing it by $D(p)$ does not compute the four-term phase. In particular, a formula for its symmetric part using only $u'(y)$ at the two external points, with no subtracted endpoint response, is not a consequence of (18). The direct phase test in `frame_edge.py` retains both endpoints. **(d)** No claimed Phase 2i symmetric-limit formula is used to extrapolate these data.

6 Why there is still no unique one-loop phase prediction

An explicit remaining freedom (c) (`frame_identities.py`, `frame_analysis.py`). For $y_1, y_2 > 0$ and any real λ, define

$$A_\lambda(y_1,y_2) = \lambda \frac{y_1 y_2 (y_1 - y_2)}{(y_1 + y_2)^3}. \tag{23}$$

It is antisymmetric, vanishes if either momentum label goes to zero with the other fixed, and has a nonzero mixed derivative. Replacing a candidate frame representative B by $B+hA_\lambda$ changes no one-body momentum, energy, Jacobian, classical action limit, or symmetric reciprocity identity. Nevertheless it changes the phase at order h^0, with

$$A_\lambda(Y, r^2Y) = \lambda \frac{r^2(1-r^2)}{(1+r^2)^3}, \qquad 0 < r < 1, \tag{24}$$

independently of Y. A mixed rectangle of this term is nonzero, so this freedom is not a one-body phase convention in disguise. This is a counterexample to *uniqueness from the one-body and reciprocity constraints*; it is not a proposal to modify the microscopic Toda theory.

(d) The missing input is a microscopic derivation of the quantum canonical position/frame map (or equivalent two-particle quantization/delay condition) that fixes the antisymmetric part of the transformation to the continuum Darboux frame. It must identify the quantum compression and momentum charges including endpoint variations. Exact one-body dispersion, or imposing scalar antisymmetry, does not supply it. Deriving that input, together with a uniform large-Y edge analysis, is required before a continuum one-loop calculation can be assigned a unique mandatory Toda target.

Consequently no number or table in this note is advertised as the requested physical two-soliton one-loop phase. **(d)** A continuum calculation using the same inherited coordinates can compare to the one-body dispersion and must respect the exact momentum Jacobian; its full phase cannot yet be tested against a uniquely fixed scalar target. This is the precise obstruction in this partial result.

7 Limits and the conditional continuum comparison

What commutes and what is not established. **(c)** At finite h, within a smooth invertible sea problem with $D > 0$, inversion from p to P is a regular coordinate change: the chain rules above and endpoint referencing can be performed exactly in either coordinate. In the displayed ambiguity example, $h \to 0$ and $Y \to \infty$ commute at fixed r, since (24) is independent of both. That observation is *only* about the counterexample. **(d)** Commutation of $h \to 0$ and $Y \to \infty$ for the full physical phase has not been established. At fixed h, $Y \to \infty$ exits the small-amplitude corner, since $\kappa^2 = hY/b$; a simultaneous small-amplitude limit needs $hY/b \to 0$. Taking $Y \to \infty$ first is not the same continuum procedure. **(c)** Likewise $r \to 0$ at fixed Y takes the smaller excitation to its edge, whereas a semiclassical expansion of both excitations requires $r^2Y \gg 1$. Equal amplitudes require separate analysis: the classical logarithm $\log[(1-r)/(1+r)]$ is singular and the spectral separation need not be large when $(1-r^2)Y$ is small. These facts obstruct a uniform termwise expansion; they are not proofs that the exact physical amplitudes have unequal iterated limits. **(d)** The actual phase limit-commutation questions at $r = 0, 1$ remain open in this work, including the head-on channel.

Comparison with `oneloop_phase.tex`. **(c)** Conditional on that note's mode-sum momentum shift, $\Delta P_g = \epsilon\alpha(\log\alpha + c)/\pi + O(\alpha^3 \log\alpha)$, $c = -\log 2 - 1 + \frac{1}{4}\log 3$, the leading relative shift is

$$\frac{\hbar_B \Delta P_g}{P_{g,0}} = \frac{3}{2\pi y}(\log\alpha + c) + \cdots, \qquad y = \alpha^2/\hbar_B. \tag{25}$$

The algebra is independently checked in `frame_identities.py`. The expansion is nonuniform in the fixed-y edge limit; it requires $y \gg |\log\alpha|$. The master report also makes the physical one-loop determinant, endpoint domain, and normalization conditional. **(d)** Neither reversing the overtaking ordering nor replacing that momentum shift by the proposed collective one proves a common quantum dictionary. We therefore do not claim agreement or inequivalence with the continuum phase integrals. No new regularization or absolute phase convention is selected.

8 Verification, error budget, and failures recorded

Numerical protocol (c) . `frame_edge.py` is a fresh finite-sea Nyström implementation using SciPy digamma and log-gamma functions, independent of the old solver classes. It solves the count equation at fixed $a = 1$ and uses LU reuse for backflow. The base meshes are $N = 325, 450, 700, 1200, 2200$ at $h = .08, .04, .02, .01, .005$; refinements have $N = 1620$ at .01 and 2750 at .005. The classical momentum target is inverted before every energy or phase measurement. No result is obtained by fitting to a proposed target function. `frame_identities.py` independently uses mpmath at 60 digits, $N = 40, 64$, $h = .4$, $q = 1.5$, and determines a from counting. It tests energy and momentum derivatives, endpoint identities, parity samples, and pressure responses with symmetric q differences of step 10^{-10}. Maximum tested errors are approximately 1.31×10^{-9} and 4.62×10^{-15}, respectively; quadrature and finite-difference errors, not decimal precision, limit these tests.

The central edge fit is quadratic through the last three h values. The extrapolation spread is the maximum difference from a last-two linear fit, a last-four cubic fit, and a last-four fit in $1, h, h^2, h \log h$. The mesh column is the largest observed refined-minus-base difference at .01 or .005. The JSON also multiplies this by the largest ℓ^1 norm of the extrapolation weights as a conservative diagnostic of amplification. None of these is a rigorous confidence interval or bound on omitted terms. Large-Y fit-model spread is separate from edge-extrapolation spread. There is no assigned numerical error bar for the unknown physical frame.

Phase 2i use policy. (c) The only checked Phase 2i formula used for a diagnostic is the product of its printed factors. Direct 60-digit Gamma-function evaluation at $|\omega| = .2, 1, 3$ gives

$$\Phi_{\mathrm{WH}}(i\omega)\Phi_{\mathrm{WH}}(-i\omega) = \frac{\pi e^{|\omega|/2}}{\sinh(|\omega|/2)} \tag{26}$$

to better than 10^{-55}, and disagrees with its printed Fourier symbol. This verifies a factor product only; zero-frequency distributional issues, the half-line inverse, $c_{\mathrm{edge}}, d_0, e_0$, Krein convolution, and $c_1(r)$ are not adopted. The new numerical dispersion and remainder data do not use any of those formulas.

Claim	Label	Check or limitation
Exact energy/backflow identity and velocity	(c)	`frame_identities.py`; independent `frame_edge.py`
Physical-momentum inversion and quantum Jacobian	(c)	symbolic chain rule; direct solve
Pressure interpretation of endpoint responses	(c)	60-digit finite differences
One-body fixed-P energy and finite-y tables	(c)	mesh and extrapolation diagnostics
Nonzero mixed edge remainder at tested Y	(c)	direct reciprocity and rectangles
Remaining antisymmetric freedom	(c)	symbolic counterexample; mixed rectangle
Unique scalar one-loop phase; full commuting limits	(d)	not established

Failures recorded.

1. The hoped-for disappearance of the classical-frame mixed remainder under the physical-momentum inversion is not supported by the new edge scan. This negative result is retained, not removed by redefining a reference.

2. The printed Phase 2i Fourier symbol fails the independent factor-product check. No other Phase 2i result is promoted to an input on that basis.

3. The attempted determination of a unique scalar one-loop target fails: (23) preserves all imposed one-body and reciprocity constraints but changes that target. This is an obstruction of the stated derivation, not a proof that a microscopic answer does not exist.

4. Large-Y extrapolation and the endpoint limits are unresolved; the numerical error diagnostics must not be presented as a proof of a constant asymptote or of commutation of exact limits.

5. The first LaTeX build failed because its temporary output directory had not yet been created (no log file could be opened). Creating that directory fixed the build setup; this was not a sandbox rejection. The first successful layout also had two overlong displayed equations; both were split before the final rendering. One render was started while a rebuild was still writing the PDF and failed with an empty-stream error; it was repeated after the build completed.

6. No numerical assertion tolerance was weakened. No existing project script or document was modified. The full historical suite was not rerun: these deliverables contain independent checks, and no prior implementation is changed or newly certified.

Reproduction. Run with one BLAS thread from `phase2/scripts/`:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 frame_edge.py --recompute
python3 frame_identities.py
python3 frame_analysis.py
```

Without `--recompute`, the edge script validates its saved JSON. Every completed checker ends with `ALL CHECKS PASSED`; this certifies its explicitly stated identities and diagnostics, not the unresolved physical identifications. SHA-256 provenance and raw timings are saved in the JSON outputs. Build from `phase2/` with `pdflatex` twice. The recorded full edge scan takes about 326 seconds, the high-precision identity run about 178 seconds, and the cached analysis under one second on this machine. A complete regeneration is therefore about eight and a half minutes; cached edge validation avoids the dense solves.